MARMARA UNIVERSITY
COMPUTER SCIENCE ENGINEERING
LINEAR ALGEBRA
FINAL HOMEWORK
22/06/2020

Write your solutions on a standard white paper. You should write your name, last name, student number, the declaration of truth (DOĞRULUK BEYANI: Bu sınav kağıdındaki tüm sorular tarafımca çözülmüş olup, hiçbir yolla başka bir şahıstan yardım alınmamış ve başka bir şahısla paylaşılmamıştır.) and your signature on each paper. Scan your solution papers possibly with your phone and upload it to ues.marmara.edu.tr as a single pdf file with file name name_lastname before 13:00. If you encounter upload problems, you can send your solution pdf file to the email address taylan.sengul@marmara.edu.tr. Each problem is worth 20pts. Show your solution steps clearly to get credit. Good luck!

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1. Are the following subsets subspaces of $\mathbb{R}^3$? If so, find their basis. If not, explain the reason.
(A) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : xyz = 0 \right\}$, (B) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x \le y \le z \right\}$, (C) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x = y \right\}$, (D) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x = 1 \right\}$.

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2. For what values of h, do the vectors $\begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} -6 \\ 3 \\ h \end{pmatrix}$ span $\mathbb{R}^3$?

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3. Is there a linear transformation $T : \mathbb{R}^2 \mapsto P_3$ which maps $T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = x^3 - 1$, and $T\begin{pmatrix} -2 \\ 1 \end{pmatrix} = x^2 + x + 1$? If there is, find $T\begin{pmatrix} 7 \\ 4 \end{pmatrix}$.

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4. Let
$$A = \begin{bmatrix} 1 & -10 & -24 & -42 \\ 1 & -8 & -18 & -32 \\ -2 & 20 & 51 & 87 \end{bmatrix}$$
Find a basis for the solution space of $A\mathbf{x} = \mathbf{0}$.

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5. (A) Let A be a square matrix. If λ_1 is an eigenvalue of A with eigenvector v_1, and λ_2 is an eigenvalue of A with eigenvector v_2, show that $A^n(c_1 v_1 + c_2 v_2) = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$ for any positive integer n and any scalars c_1, c_2. (Here A^n is the matrix A multiplied n times with itself.)
(B) Suppose $\lambda_1 = 2$, $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\lambda_2 = 1$ and $v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ above. Find $A^{100}\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.